

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 1

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO1

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 1

Unit Title: Foundations of Unsupervised Learning

Unit Mapped RBT: L2 (Understand)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Identify the measure that gives the straight-line gap between two records.

- A) Euclidean distance
- B) Manhattan distance
- C) Cosine similarity
- D) Cosine distance

Q2. State the range of values cosine similarity can take when the vectors may hold negative entries.

- A) From 0 to 1 only
- B) From -1 to +1
- C) From 0 to infinity
- D) From -infinity to +infinity

Q3. Define cosine distance in terms of cosine similarity.

- A) The reciprocal of the cosine similarity
- B) The square of the cosine similarity
- C) One minus the cosine similarity
- D) The negative of the cosine similarity

Q4. Name the stage of an analysis in which the data is summarised and plotted before any model is built.

- A) Model deployment
- B) Hyperparameter tuning
- C) Cross validation
- D) Exploratory data analysis

Q5. Identify what a clustering algorithm produces for each record of the input table.

- A) A predicted numeric value
- B) A cluster label
- C) A probability of fraud
- D) A confidence interval

Q6. Name the unsupervised task that finds tightly connected groups of users inside a social network.

- A) Sentiment analysis
- B) Link prediction
- C) Spam classification
- D) Community detection

Q7. Define semi-supervised learning.

- A) Learning from a few labelled records and many unlabelled ones
- B) Learning from labelled records only
- C) Learning from a reward given after each action
- D) Learning from records that have two target columns

Q8. State the value a Euclidean distance takes when the two points are identical.

- A) One
- B) Infinity
- C) Zero
- D) It is undefined

Q9. Name the three measures listed on the syllabus for comparing a pair of records.

- A) Inertia, silhouette and the Davies-Bouldin Index
- B) Support, confidence and lift
- C) Euclidean distance, Manhattan distance and cosine similarity
- D) Accuracy, precision and recall

Q10. Name the quantity that fixes the number of dimensions of the space in which the records lie.

- A) The number of features
- B) The number of records
- C) The number of clusters
- D) The number of iterations

Q11. State what the target column of a supervised dataset holds.

- A) The row number of each record
- B) The distance to the nearest record
- C) The cluster each record belongs to
- D) The correct answer for each record

Q12. Identify the operation Euclidean distance applies to the sum of the squared differences.

- A) The absolute value
- B) The square root
- C) The natural logarithm
- D) The reciprocal

Q13. Name the type of feature that records a category rather than a number.

- A) A continuous feature
- B) A derived feature
- C) A target feature
- D) A categorical feature

Q14. Recall what knowledge discovery in an unlabelled dataset sets out to produce.

- A) A forecast of next month's revenue
- B) A label for every record in the table
- C) Patterns that were not known in advance
- D) A ranking of the features by importance

Q15. State what must be chosen before any distance-based unsupervised method can be run.

- A) A target column to predict
- B) A measure of distance between records
- C) A training and a test partition
- D) A confidence level for the result

Q16. Identify the term used for a single row of an unlabelled dataset.

- A) A record
- B) A cluster
- C) A centroid
- D) A feature

Q17. Explain why cosine similarity is unsuitable when the total amount a customer spends is the quantity of interest.

- A) It cannot be computed on spending data
- B) It ignores the size of the vectors entirely
- C) It always returns a value close to zero
- D) It requires the number of groups in advance

Q18. Distinguish a clustering task from a classification task by what each one is given.

- A) Clustering is given no labels, classification is given labels
- B) Clustering is given labels, classification is given none
- C) Both of the two are given a full set of labels
- D) Both of the two are given a single labelled record

Q19. Predict the effect on a Manhattan distance of adding one extra feature whose value is identical in both records.

- A) The distance is doubled
- B) The distance falls to zero
- C) The distance can no longer be computed
- D) The distance is left unchanged

Q20. Interpret a cosine similarity of exactly zero between two purchase vectors holding non-negative counts.

- A) The two customers bought identical baskets
- B) The two customers spent the same amount
- C) The two customers bought no item in common
- D) One of the two customers bought nothing

Q21. Predict what happens to a Euclidean distance when forty uninformative random features are added to a table.

- A) The genuine differences are drowned out
- B) The genuine differences are sharpened
- C) The distance becomes exactly zero
- D) The distance stops depending on the data

Q22. Recognize the unsupervised task described: a telecom operator groups its cell towers by their hourly

traffic profile so that capacity can be planned tower by tower.

- A) Supervised regression on traffic
- B) Classification of tower faults
- C) Clustering for segmentation
- D) Reinforcement learning of a policy

Q23. Summarize why labels are frequently unavailable in a real project.

- A) Labels are forbidden by data protection law
- B) Labelling is slow, costly and often needs an expert
- C) Labels cannot be stored in a numeric table
- D) Labels make a model less accurate overall

Q24. Explain why two customers with identical spending patterns but very different totals sit far apart under Euclidean distance.

- A) Euclidean distance only compares the largest feature
- B) Euclidean distance cannot handle spending data
- C) Euclidean distance ignores the smaller of the two
- D) Euclidean distance grows with the size of the values

Q25. Distinguish a Euclidean distance of zero from a cosine similarity of zero.

- A) Zero distance means unrelated, zero cosine means identical
- B) Both of the two mean that the records are identical
- C) Zero distance means identical, zero cosine means unrelated
- D) Both of the two mean that the records are unrelated

Q26. Interpret a cosine similarity of -1 between two feature vectors.

- A) They are identical to one another
- B) They share no feature in common
- C) One of the two is the zero vector
- D) They point in exactly opposite directions

Q27. Explain the difficulty a categorical feature introduces into a Euclidean distance.

- A) Every pair of different categories ends up equally far apart
- B) The distance can no longer be computed at all
- C) The distance becomes negative for some of the pairs
- D) The two categories are silently merged into one

Q28. Predict the effect on a clustering of removing the one feature that carried the real structure.

- A) The groups become sharper than before
- B) The groups it revealed can no longer be found

- C) The algorithm refuses to run at all
- D) The number of groups doubles automatically

Q29. Describe what the formulation of an unsupervised problem has to settle.

- A) What accuracy the model must reach
- B) How many records are held back for testing
- C) Which label is attached to each record
- D) What is measured and what counts as similar

Q30. Recognize the situation in which semi-supervised learning is the natural choice.

- A) Every record carries a reliable label
- B) A few records are labelled and very many are not
- C) No record can ever be labelled at all
- D) Every record carries two different labels

Q31. Explain why the result of an unsupervised method has to be reviewed by someone who knows the domain.

- A) The algorithm cannot compute the distances reliably
- B) The software licence requires a human reviewer
- C) There is no ground truth to check the result against
- D) The records must be relabelled before reporting

Q32. Distinguish exploratory data analysis from model building.

- A) Exploration describes the data, modelling acts on it
- B) Exploration predicts values, modelling plots them
- C) Both of the two produce a fitted model
- D) Both of the two require a labelled target

Q33. Explain why Manhattan distance is also called city-block distance.

- A) Movement is measured along the axes, not diagonally
- B) It can only be used on map coordinates
- C) It was first devised for planning cities
- D) It measures the straight line between points

Q34. Explain why fraud detection is often treated as an unsupervised problem even when a few past frauds are on record.

- A) Past fraud records are legally unusable
- B) Fraud records cannot be stored numerically
- C) A labelled model would run far too slowly
- D) New fraud patterns rarely resemble the recorded ones

Q35. Predict what a clustering algorithm reports on a dataset that contains no genuine group structure.

- A) It reports that no clusters were found

- B) It still returns the number of groups it was asked for
- C) It refuses to run and raises an error
- D) It returns every record as its own cluster

Q36. Recognize the fault described: an analyst clusters a customer table in which the postcode is stored as a plain number, and obtains groups that follow numeric closeness in that column.

- A) The number of clusters was set too high
- B) The distance measure was chosen too late
- C) A categorical code has been treated as a numeric quantity
- D) The table held too few records to cluster

Q37. Interpret a large Euclidean distance between two customers who purchased exactly the same set of products.

- A) They have no product in common at all
- B) One of the two bought nothing whatever
- C) They differ in quantity rather than in what they bought
- D) The distance has been computed incorrectly

Q38. Summarize why the choice of distance can matter more than the choice of algorithm.

- A) The algorithm is chosen automatically by the software
- B) The algorithm sees the data only through the distances
- C) Every algorithm produces exactly the same groups
- D) The distance decides how many records are kept

Q39. Explain why a similarity measure and a distance measure move in opposite directions.

- A) A similarity is always negative in value
- B) A distance can never exceed the value one
- C) A similarity is computed on the labels only
- D) Records that are near are the records that are alike

Q40. Predict the effect on a market segmentation of including the customer's account number as a feature.

- A) It adds meaningless variation that can distort the groups
- B) It sharpens the boundaries between the groups
- C) It guarantees that every group is the same size
- D) It has no effect of any kind on the result

----- End of Sample Question Set -----